

Marketplace Seller Performance Analysis

Improving Marketplace Performance Through Seller, Delivery, and Customer
Behavior Analytics

Business Report

Based on the Olist Brazilian E-Commerce Public Dataset

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Summary

This report analyzes marketplace performance across seller behavior, delivery reliability, customer satisfaction, revenue drivers, and regional patterns, using ~100,000 real orders from the Olist marketplace (2016-2018). The analysis combines SQL-based data cleaning, Python-based exploratory analysis, and an interactive Power BI dashboard to move from raw transactional data to concrete operational recommendations.

Three findings stand out as priorities for leadership attention: a small set of high-volume sellers combine strong revenue with poor delivery and review performance; delivery delay is the single strongest driver of customer dissatisfaction in the dataset; and revenue is heavily concentrated in São Paulo while delivery reliability is weakest in the Northeast, including one high-revenue state (Rio de Janeiro) that underperforms on delivery despite its commercial importance.

Methodology

Nine relational tables (orders, order items, customers, sellers, products, payments, reviews, geolocation, and category translations) were loaded into a SQLite database and joined into a single order-item-level master table (112,650 rows). A data quality check identified and resolved a duplicate-review join bug that would have inflated row counts by ~0.6%, corrected using row-level deduplication to keep only the most recent review per order. Delivery-time analysis was limited to orders with status "delivered" (97% of all orders), since incomplete orders lack delivery timestamps by definition. All monetary figures throughout this report are in Brazilian Real (R\$), the native currency of the dataset.

Key Findings

1. Seller Performance & Risk

Revenue leadership among sellers follows two distinct patterns: high-volume sellers (1,000+ orders at moderate price points) and high-ticket sellers (fewer orders at higher average value). The top seller generated approximately R\$229,000 in revenue across 1,132 orders.

One seller was identified with a 92% late-delivery rate across 25 completed orders — a genuine, statistically meaningful outlier rather than a small-sample artifact, and a clear candidate for audit.

Across all sellers, order volume and late-delivery rate show almost no correlation ($r = 0.01$). Delivery risk is not concentrated among small or new sellers; it is spread broadly, meaning delivery monitoring should apply uniformly rather than focusing only on smaller sellers.

A combined view of revenue, review score, and delivery risk identified ten sellers with meaningful order volume (50+ orders) and consistently poor reviews (under a 3.1 average) — the priority list for seller support or removal.

2. Customer Experience

Average review score declines sharply and consistently with delivery delay: 4.21 for on-time or early orders, 2.68 for orders 1-7 days late, and 1.70 for orders more than 7 days late. A correlation

analysis confirms this relationship ($r = -0.23$ between delay and review score) and rules out price or freight cost as meaningful drivers of satisfaction (both near-zero correlation with review score).

The lowest-rated product categories are not explained by delivery delay: `security_and_services` (2.50 average review score) is delivered 17.0 days early on average, and `diapers_and_hygiene` (3.38 average review score) 11.4 days early — both earlier than the -12.1 day dataset average. Since even the most reliably early-delivered category has the worst review score, this rules out logistics as the driver and points to product or category-specific issues instead — a distinction that should shape how each category is addressed.

3. Revenue & Cost Efficiency

`Christmas_supplies` and `signaling_and_security` carry the highest freight cost relative to revenue (30-37%), representing a logistics efficiency concern independent of raw profit margin, which cannot be directly calculated from this dataset due to the absence of seller-side cost data.

Credit card is the dominant payment method (73.8% of orders), followed by boleto, a Brazil-specific bank-slip payment method (19.4%). Average order value rises steadily with installment count, from approximately R\$91 (1 installment) to R\$292 (10 installments), reflecting expected customer behavior of splitting larger purchases.

4. Regional Performance

Revenue is heavily concentrated in São Paulo, which generates approximately R\$5.2 million — more than three times the next-highest state, Rio de Janeiro (~R\$1.8 million).

Delivery reliability is weakest in Brazil's Northeast (Alagoas, Maranhão, Sergipe, Ceará, and Piauí), with late-delivery rates between 15% and 22%, likely reflecting distance from the São Paulo-centered seller base. Rio de Janeiro, despite being the second-highest revenue state, also shows an above-average late-delivery rate (13.3%) — a flag worth addressing given its commercial importance.

Limitations

This analysis has no seller-side cost data, so all cost-efficiency findings are framed as freight-to-revenue ratios rather than true profit margin. The dataset's final month (September 2018) contains only a single order and was excluded from time-trend analysis as an artifact of the data collection window rather than a real business event. Findings reflect historical patterns (2016-2018) and would benefit from validation against current data before being acted on operationally.

Conclusion

The combination of seller-level risk scoring, a confirmed delay-satisfaction relationship, and a regional reliability gap gives Olist's leadership a concrete, prioritized starting point: audit the identified high-risk sellers, address delivery reliability in the Northeast and Rio de Janeiro, and separate product-quality issues from logistics issues when addressing low-rated categories. The accompanying Power BI dashboard provides an ongoing, interactive view of these metrics for continued monitoring.